

1) Define LTI-systems. Explain with examples, how periodic and non-periodic signals are distinguished.

→ LTI System stands for Linear Time Invariant System.

→ A system is called LTI system if it satisfies

a) Linearity

b) Time invariance.

a) Linearity:-

A system is linear if it obeys the principles of Additivity & Homogeneity

for additivity & Homogeneity.

if $x_1(t) \rightarrow y_1(t)$

$x_2(t) \rightarrow y_2(t)$

then,

$$a x_1(t) + b x_2(t) \rightarrow a y_1(t) + b y_2(t)$$

where a and b are constants.

b) Time invariance.

A system is time invariance if shifting the input signal causes the same shift in the output signal.

here, if $x(t) \rightarrow y(t)$

then,

$$x(t - t_0) \rightarrow y(t - t_0)$$

for any time shift t_0 .

Example of LTI System

$$y(t) = x(t) + x(t-1)$$

This system is linear & time invariant.

periodic and Non-periodic signal:-

- periodic signal:-

A signal is called periodic if it repeats itself after a fixed interval of time.

Mathematically,

$$x(t) = x(t+T)$$

where T is the fundamental period.

eg:-

$$x(t) = \sin(2\pi t)$$

This signal repeats after every 1 second, so it is periodic.

- Non-periodic signal

A signal which does not repeat itself after any interval of time is called non periodic signal.

eg:-

$$x(t) = e^{-t} u(t).$$

→ this signal decreases continuously & never repeats.

⇒ Periodic signal

- A signal that repeats itself after a fixed interval of time

- Has a fundamental period T

- Exists continuously in a repeating pattern

- usually power signals.

- eg:- sine wave / cosine wave

Non-periodic signal.

- A signal that does not repeat itself after any interval of time.

- Does not have a fundamental period

- Exists without repetition.

- usually energy signals.

- eg:- exponential decay signal, pulse signal.

Q. Define Energy and power signal.

Energy signal:-

A signal is called an energy signal if:

- Its total energy is finite
- Its average power is zero.

For a continuous-time signal,

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$\text{if } 0 < E < \infty$$

then the signal is called an energy signal.

eg:- $x(t) = e^{-t} \cdot u(t)$.

This signal has finite energy, so it is an energy signal.

power signal

A signal is called a power signal if:

- Its average power is finite
- Its energy is infinite.

For a continuous-time signal,

$$P = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

$$\text{if } 0 < P < \infty$$

then the signal is called a power signal.

eg $x(t) = A \cos(\omega t)$.

Energy signal

- A signal having finite energy is called an energy signal
- Total energy is finite
- Average power is zero
- Generally exist for finite duration
- eg:- pulse signal, exponential signal

Power signal

- A signal having finite average power is called power signal
- Total energy is infinite
- Average power is finite & non-zero
- Generally exists for infinite duration
- eg:- sinusoidal signal

3 → How do you classify the signal?

→ signals can be classified based on various characteristics and properties. The major classification schemes are:-

1 → Based on Time Domain (Continuous-time vs Discrete-time).

1A → Continuous-Time Signals (Analog signals).

- Signals are defined for all values of time,
 - Denoted as $x(t)$ where t is a continuous variable.
 - can take any value in a given range.
- eg:-

Audio signals.

Temperature variation $T(t) = 25 + 5 \sin(2\pi t)$.

B) Discrete Time Signals (Digital Signals).

- Signals defined only at discrete point in time.
- Denoted as $x[n]$ where n is an integer ($n = \dots -2, -1, 0, 1, 2, \dots$)
- Result of sampling continuous signals at regular intervals.

eg:-

- Digital audio (44.1 kHz sampling).
- population census data
- $x[n] = 5 \sin(n\pi)$

2) Based on periodicity (Periodic vs Non-Periodic)

A) Periodic Signals.

- Signals that repeats themselves after a fixed time interval called the period (T)
- Mathematically, $x(t) = x(t+T)$ for all t , where $T > 0$ is the fundamental period.
- For discrete signals $x[n] = x[n+N]$ where N is the fundamental period in samples.

eg:- $\sin(\omega t)$, $\cos(\omega t)$

B) Non-Periodic Signals (Aperiodic).

- Signals that do not repeat themselves.
- Do not satisfy $x(t) = x(t+T)$ for any finite $T > 0$
- Most real-world signals are non-periodic.

eg:-

Speech signal, e^{-t} , Random noise, $u(t)$.

3) Based on Energy And power

A) Energy signal:-

- have finite energy and zero average power

- Energy: $E = \int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$

- power: $p = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt = 0$

- It is typically non-periodic, finite duration signals.

eg:- e^{-t} , $x(t) = A[u(t+1) - u(t-1)] \rightarrow$ pulse

B) power signal:-

- have infinite energy but finite average power.

- Energy $E = \infty$

- power $p = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt < \infty$

- It is typically periodic or infinite-duration signals

eg:- $x(t) = C$, square wave, $x(t) = A \cos(\omega t)$

4) Based on Casuality (Casual vs Non-Casual)

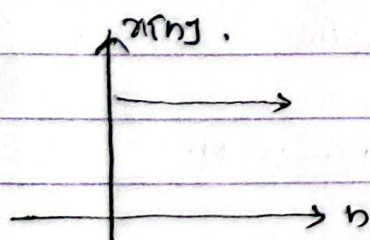
A) Casual signals

- Zero for all $t < 0$ or $n < 0$

- Depend only on present & past values.

eg

$x(t) = e^{-2t} \cdot u(t)$, $x(t) = u(t-1)$



B) Non-Causal signals.

- Non-zero for $t < 0$ or $n < 0$.
- Depends on future values.

eg:- $x(t) = e^{2t} u[-t]$.

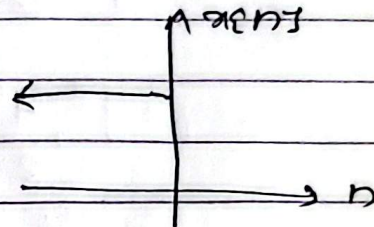


😊 A non-causal signal exists in positive as well as negative time.

ignore 😊.

X (C) Anti-causal signal

⇒ When the non-causal signal is defined only for $n \leq 0$, then it is called Anti-causal signal.



⇒ Anti-causal signal does not exist for positive time

$$x[n] = \{1, -1, 2, 3, 4\}$$

4.) Based on Symmetry (Even vs Odd).

A. Even signal.

- $x(-t) = x(t)$.
 - Symmetric about the vertical axis.
- eg:- $\cos(\omega t)$, $x(t) = |t|$

B. Odd signal

- $x(-t) = -x(t)$, or
- Antisymmetric about the vertical axis.

eg:- $\sin \omega t$
 $x(t) = t$

Q. Check whether $x(t) = A \cos \omega t$ is a power or energy signal.

→ here, $x(t) = A \cos \omega t$.

for
To check energy signal,

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$= \int_{-\infty}^{\infty} A^2 \cos^2 \omega t \cdot dt$$

$$= \frac{A^2}{2} \int_{-\infty}^{\infty} (\cos 2\omega t + 1) dt$$

$$= \frac{A^2}{2} \left[\frac{\sin 2\omega t}{2\omega} + t \right]_{-\infty}^{\infty}$$

$$= \infty$$

Energy is infinite

Now, checking for power

$$P = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

$$= \lim_{T \rightarrow \infty} \frac{A^2}{2T} \int_{-T}^T \cos^2 \omega t \cdot dt$$

$$= \lim_{T \rightarrow \infty} \frac{A^2}{4T} \int_{-T}^T \frac{\cos 2\omega t + 1}{2} dt$$

$$= \lim_{T \rightarrow \infty} \frac{A^2}{4T} \left[\frac{\sin 2\omega t}{2\omega} + t \right]_{-T}^T$$

$$= \lim_{T \rightarrow \infty} \frac{A^2}{4T} \left[\frac{\sin 2\omega T}{2\omega} + \frac{\sin 2\omega T}{2\omega} + T + T \right]$$

$$= \lim_{T \rightarrow \infty} \frac{A^2}{4T} \left[\frac{2 \sin 2\omega T}{2\omega} + 2T \right]$$

we know, $\omega T = 2\pi$

$$= \lim_{T \rightarrow \infty} \frac{A^2}{4T} \left[\frac{2 \sin 4\pi}{4\pi} + 2T \right]$$

$$= \lim_{T \rightarrow \infty} \frac{A^2}{4T} \times 2T$$

$$\therefore \frac{A^2}{2}$$

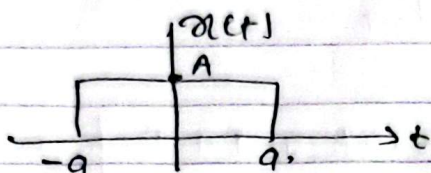
here, Avg power = $\frac{A^2}{2}$ finite

So, energy is ∞ & Avg power is finite,

$\therefore$ It is power signal.

5) Determine whether the signal $x(t) = A [u(t+a) - u(t-a)]$ for $a > 0$ is a power or energy signal.

→ Given



$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$= \int_{-a}^a A^2 dt$$

$$= A^2 [t]_{-a}^a$$

$$E = 2aA^2, \text{ finite}$$

$$P = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T A^2 dt$$

$$= \lim_{T \rightarrow \infty} \frac{A^2 [2a]}{2T}$$

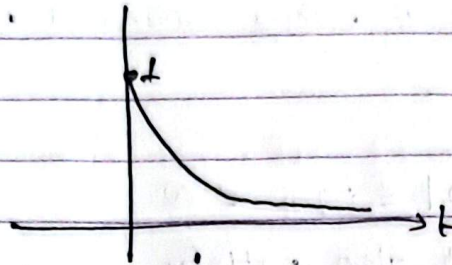
$$= 0$$

here, energy is finite & avg power is zero
so the signal is energy signal.

Q. Find the energy and power of signal $x(t) = e^{-4t} u(t)$ & justify whether it is energy signal or power signal.

→ soln

$$x(t) = e^{-4t} u(t)$$



for energy

$$E = \int_{-\infty}^{\infty} e^{-8t} dt$$

$$= \left[\frac{e^{-8t}}{-8} \right]_0^{\infty}$$

$$= \frac{1}{8} [e^{-\infty} - e^0]$$

$$\therefore E = \frac{1}{8}$$

for power,

$$P = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x^2(t) dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_0^T e^{-8t} dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{2T} \left[\frac{e^{-8t}}{-8} \right]_0^T$$

$$= \lim_{T \rightarrow \infty} \frac{1}{2T \times -8} [e^{-8T} - e^0]$$

$$= 0$$

finite energy & Avg power = 0 so it is energy signal.

7) How would you say that a system is BIBO stable?

Comment about the stability for the following system

$$y(n) = 2x(n+1) + [x(n-1)]^2.$$

→ A system is said to be BIBO stable (Bounded Input Bounded output) stable if every bounded input produces a bounded output.

if $|x(n)| \leq M_x < \infty$
then output must also satisfy

$$|y(n)| \leq M_y < \infty$$

for some finite value M_y .

→ Given system,

$$y(n) = 2x(n+1) + [x(n-1)]^2$$

Assume Input is bounded.

$$|x(n)| < M$$

then,

$$|2x(n+1)| \leq 2M$$

$$\& \quad |[x(n-1)]^2| \leq M^2$$

therefore,

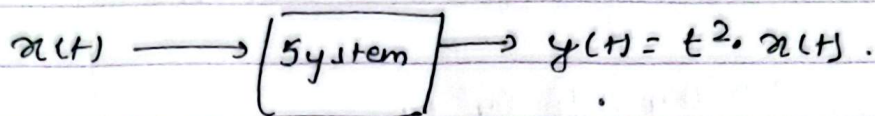
$$|y(n)| \leq 2M + M^2$$

which is finite

so comment about the linearity, stability, time-invariance, & causality for the following system: $y(t) = t^2 x(t)$.

→ Given, $y(t) = t^2 x(t)$.

① Linearity



$$x_1(t)$$

$$y_1(t) = t^2 \cdot x_1(t)$$

$$x_2(t)$$

$$y_2(t) = t^2 \cdot x_2(t)$$

$$y_3(t) = A \cdot t^2 x_1(t) + B \cdot t^2 x_2(t)$$

$$A x_1(t) + B x_2(t) \rightarrow \boxed{\text{System}} \rightarrow y_4(t) = t^2 [A x_1(t) + B x_2(t)]$$

here, $y_3(t) = y_4(t)$, hence system is linear system.

② stability

$$y(t) = t^2 x(t)$$

let have a finite input $x(t) = 1$,

then

$$y(t) = t^2 \quad \text{at } x(t) = 1,$$

even with the bounded i/p the output is not finite
so the system is unstable.

Since 😊 As $t \rightarrow \infty$, $y(t) \rightarrow \infty$.

③ Time Invariance.

$$y(t) = t^2 \cdot x(t).$$

→ Delay i/p by t_1 , $x(t-t_1)$.

$$y(t, t_1) = t^2 \cdot x(t-t_1)$$

→ Delay o/p by t_1

$$y(t-t_1) = (t-t_1)^2 \cdot x(t-t_1)$$

Since,

$$y(t, t_1) \neq y(t-t_1) \Rightarrow \text{Time variant.}$$

④ Casuality.

→ If the output of system depends on present i/p $x(t)$ & past input only then such system is called casual system.

here, $y(t) \rightarrow$ depends on $x(t)$.

present i/p only so

the system is casual.

⑤ Check casuality and time-invariance of a signal.

$$y(t) = x(t) + u(t-5).$$

→ Given signal/system.

$$y(t) = x(t) + u(t-5)$$

here

$u(t-5)$ is a fixed signal.

& the output only depends on present i/p $x(t)$ so the given system is casual

$$\rightarrow x(t) \rightarrow \boxed{\text{System}} \rightarrow x(t) + y(t-s)$$

Delay t_1 in input, $x(t-t_1)$.

$$y(t, t_1) = x(t-t_1) + y(t-s)$$

Delay t_1 in o/p.

$$y(t-t_1) = x(t-t_1) + y(t-t_1-s)$$

Since, $y(t, t_1) \neq y(t-t_1)$ so system is
Time variant

(15) Check whether the given system is linear, casual & time invariant or not.

$$1) y(t) = \log x(t)$$

$$x(t) \rightarrow \boxed{\text{System}} \rightarrow y(t) = \log x(t)$$

$$x_1(t) \rightarrow y_1(t) = \log x_1(t)$$

$$x_2(t) \rightarrow y_2(t) = \log x_2(t)$$

$$y_3(t) = A \log x_1(t) + B \log x_2(t)$$

$$A x_1(t) + B x_2(t) \rightarrow \boxed{\text{System}} \rightarrow y_4(t) = \log(A x_1(t) + B x_2(t))$$

here, $y_3(t) \neq y_4(t)$

so the system is non-linear.

For Casualty

- Since the o/p depends only on the present i/p value, the system is casual

For time invariance

- Delay i/p by t_1 i.e $x(t-t_1)$

$$y(t, t_1) = \log x(t-t_1)$$

- Delay o/p by t_1 ,

$$y(t-t_1) = \log x(t-t_1)$$

Since, $y(t, t_1) = y(t-t_1)$ so the system is time invariant

② $y(t) = t \cdot x(t)$

for linearity:-

$$x(t) \rightarrow [\text{System}] \rightarrow y(t) = t \cdot x(t)$$

$$x_1(t) \rightarrow y_1(t) = t \cdot x_1(t)$$

$$x_2(t) \rightarrow y_2(t) = t \cdot x_2(t)$$

$$y_3(t) = [A x_1(t) + B x_2(t)] \cdot t$$

$$A x_1(t) + B x_2(t) \rightarrow [\text{System}] \rightarrow y_4(t) = t \cdot [A x_1(t) + B x_2(t)]$$

$$\text{Since, } y_3(t) = y_4(t)$$

For casuality,

$$y(t) = t \cdot x(t)$$

Since o/p depends only upon the present i/p value so the system is casual.

for Time Invariance,

~~The~~ Delay i/p by t_1 , i.e $x(t-t_1)$

$$y(t, t_1) = t \cdot x(t-t_1)$$

- Delay of t_1 .

$$y(t-t_1) = (t-t_1) \cdot x(t-t_1)$$

Since, $y(t, t_1) \neq y(t-t_1)$

So the system is time variant.

11) Explain why a sinusoidal wave is periodic & deterministic signal.

→ A sinusoidal signal is.

$$x(t) = A \sin(\omega t + \phi)$$

- For periodic:-

A signal is periodic if,

$$x(t) = x(t+T)$$

For some positive constant T (fundamental)

For sinusoidal wave,

$$T = \frac{2\pi}{\omega}$$

After every interval T , the waveform repeats itself

Therefore, sinusoidal wave is periodic.

- For deterministic.

- A deterministic signal can be completely described mathematically.

- Since Amplitude A , frequency ω , & phase ϕ are known, future value can be predicted exactly.

- Hence sinusoidal signal is deterministic.

12.) Find whether the signal is energy or power signal $\delta[n]$.

→ here,

$$\begin{aligned} \text{Energy} &= \sum_{n=-\infty}^{\infty} |\delta[n]|^2 \\ &= \sum_{n=-\infty}^{\infty} \{\delta[n]\}^2 \end{aligned}$$

We know, $\delta[n] = \begin{cases} 1 & \text{for } n=0 \\ 0 & \text{elsewhere.} \end{cases}$

So, $E = 1$, which is finite & non-zero.

next,

$$\text{power} = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |\delta[n]|^2$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N 1$$

$$= \lim_{N \rightarrow \infty} \frac{1}{2N+1}$$

$$= \frac{1}{\infty} = 0$$

So, Energy is finite & Avg. power is zero

So $\delta[n]$ is an energy signal.